Introductory activity

Activity 1 · What day will it be? 35 min . 17 October 2026 is a Saturday. The days of the week are numbered from 0 (Saturday) to 6 (Friday): Sunday has number 1, Monday number 2, and so on.

1. What day of the week will it be 7 days later? 14 days? 700 days?
2. What day will it be 100 days later? Explain how to get the answer without counting the days one by one.
3. Same question for 1,000 days.
4. The year 2027 is not a leap year. What day of the week will 17 October 2027 be?
5. 1 January 2027 comes 76 days after 17 October 2026. What day will it be?
6. Two durations a and b, in days, lead to the same day of the week. What can be said about $a - b$? State a general condition.

Observing is not proving: the properties conjectured here are established in the course.